

Stibitz Christin Christudas

Design & NPD Specialist | Additive Manufacturing (WAAM) Researcher
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PROFESSIONAL SUMMARY

Mechanical Engineer with 7+ years of experience across additive manufacturing research, heavy-equipment design, and industrial operations. Skilled in CAD/CAE (Creo Parametric, SolidWorks, AutoCAD, ANSYS), materials testing, and model-based definition (MBD) workflows. Published researcher in Wire Arc Additive Manufacturing (WAAM) with hands-on experience in robotic 3D printing, mechanical/material characterization, and cross-functional design collaboration for global OEMs.

CORE SKILLS

Engineering & Technical Skills: Prototyping & Design, Simulation & Analysis (FEA/CFD), Reverse Engineering, GD&T and ASTM Y14.5, Materials Testing & Characterization, Robotic Wire Arc Additive Manufacturing (WAAM), Sheet Metal Design, Welding Symbols & Standards, Hydraulic Hose & Cable Routing, 2D/3D Drafting.

Software & Tools: Creo Parametric, Creo Illustrate, Windchill & Albatross PLM, SolidWorks, AutoCAD, ANSYS (FEA, Fluid Mechanics), MATLAB, Microsoft Excel.

Professional & Soft Skills: Technical Communication, Cross-Functional Collaboration, Team Leadership, Project Management, Analytical Problem-Solving.

PROFESSIONAL EXPERIENCE

Student Researcher

Jan 2025 – Present

Georgia Southern University

Statesboro, GA, USA

- Design and execute experiments combining mechanical engineering and materials science principles for additive manufacturing applications.
- Perform robotic wire arc additive manufacturing (WAAM) 3D printing of 316L stainless steel and characterize resulting material properties for use as forging preform substitutes.
- Presented original research findings at the Annual International Solid Freeform Fabrication (SFF) Symposium, University of Texas at Austin.
- Collaborate with interdisciplinary teams on experimental design, testing protocols, and research documentation.

Senior Design Engineer

Jul 2019 – Jul 2024

UST Global (Client LIEBHERR)

Bangalore, India

Hydraulics and Electrical Design Engineering

- **533 Autonomy Truck:** Developed autonomous steering concepts, hydraulic routing, interference checks, and schematic analysis.
- **Air Heater Relocation:** Performed root-cause analysis, hose rerouting, and interference checks for air-heater relocation.
- **3D Drawings:** Led paperless-drawing initiative, developing Model-Based Assemblies (3D drawings) in Creo Illustrate from legacy 2D drawings and models, improving design traceability and review efficiency.
- **Bolt Pad Assembly:** Performed design and drawing verification for bolt pad assemblies used in Liebherr 533, 536, and 537 truck models.
- **System Design:** Hands-on experience working on CREO for mechanical and electrical integration in Frame Wiring, Traction Control Systems, and Battery Disconnect Assemblies.

Field and Structural Engineering

- **Desiccator box:** Designed a new desiccator box design along with its components to avoid condensation inside the control box.
- **Hydraulic and fuel tank sensors:** Routed new separate cables from the axle box to the Hydraulic skid and Fuel tank along with sensors.
- **Hoist cylinder protection:** Redesigned hoist cylinder covers for better protection and suspension eye changes.
- **Axle Box masks redesign:** Developed a new concept design of paint masks for the axle box.
- **Mudflaps redesign:** Redesign and development of mudflaps for DS and ODS for better protection of fuel tanks.

EARLY CAREER EXPERIENCE

Apprentice Engineer <i>Gail Gas Ltd.</i>	Jun 2018 – May 2019 <i>Bangalore, India</i>
Site / Design Engineer <i>JSE Infratech Ltd.</i>	Oct 2016 – Jul 2017 <i>Bangalore, India</i>
Summer Intern <i>Hindustan Aeronautics Ltd.</i>	Jan 2016 – Feb 2016 <i>Bangalore, India</i>

EDUCATION

Master of Science in Mechanical Engineering <i>Georgia Southern University</i>	Dec 2026 (Expected) <i>Statesboro, GA</i>
• CGPA – 3.87 • Research Focus: Additive manufacturing of metals using Wire Arc Additive Manufacturing (WAAM); material characterization of WAAM-fabricated forging preforms. • Capstone Project: Remodification of Linear scale for Howmet Aerospace.	
Bachelor of Science in Mechanical Engineering <i>Visveswaraya University</i>	May 2016 <i>Bangalore, India</i>
• Capstone Project: Modification of a 4-stroke engine into a 2-stroke air-compressed engine.	

PUBLICATIONS

- Springer – Progress in Additive Manufacturing: “Pulsed Robotic Wire Arc Additive Manufacturing of 316L Stainless Steel for Forging Preforms” – Apr 2026
- Springer – Progress in Additive Manufacturing: “From Wire Arc Additive Manufacturing (WAAM) to Forging: High-Temperature and High-Strain-Rate Deformation of WAAM-Fabricated 316L Stainless Steel” – In Progress
- SFF Symposium, UT Austin: “Pulsed Robotic Wire Arc Additive Manufacturing of 316L Stainless Steel for Forging” – Aug 2025
- SFF Symposium, UT Austin: “Evaluation of Wire Arc AM Fabricated 316L Stainless Steel Post-Forging” – Aug 2026
- SFF Symposium, UT Austin: “Wire Arc Additive Manufacturing of Layered HSLA–Stainless Steel Composite Structures for Enhanced Strength–Ductility Performance” – Aug 2026

CERTIFICATIONS & HONORS

- Lean Six Sigma Yellow Belt – Kennesaw State University
- Lean Six Sigma Green Belt – Kennesaw State University
- “Superstar” Recognition Award, UST Global – awarded 5 times for outstanding design engineering and team performance.
- Research Assistant and Teaching Assistant, Georgia Southern University.